Brain alignment as a usable training signal: the internal extended manuscript

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Abstract

For a person and a language model reading the same text, a simple linear map predicts the person's brain activity from the model's middle-layer activations. For a decade this correspondence has been used in one direction only: to measure a model's alignment with the brain after it is trained. We ask whether it can instead be used during training. We test this in the setting of model compression: we distill a language model into a smaller student with brain alignment in the objective, at a matched budget, and ask whether the student preserves or improves that alignment, a question that remains untested. An information-budget argument implies that such a signal is far too small to act as a second teacher; at most it is a weak prior. Such a prior should help only where the text signal is weakest (small models, scarce data, aggressive compression) and wash out wherever data is abundant. Under whole-story splits and a capacity-matched nuisance floor, trained representations exceed untrained controls within one deeply sampled participant, but direct brain-objective tuning shows no reliable gain at the valid fold unit. An apparent gain against a participant-averaged response disappears when nine participants become the inference unit; the tested downstream and privileged-information routes establish no practical payoff, and a learnable dense neural proxy does not transfer to improved real-brain alignment. The results make brain alignment a useful diagnostic signal under explicit controls, but not a demonstrated training signal in the regimes studied here.

Contents

Notation and terms

Encoding model

A ridge regression from a model's per-stimulus features to measured brain activity, scored out-of-sample. The standard brain-alignment instrument.

Voxel / ROI

A voxel is one 3D measurement location in an fMRI scan; an ROI is a predefined region (here, the five left-hemisphere language regions of the Tuckute ROI benchmark).

ΔR^2 (unique R^2)

The variance the language-model features add *on top of* the nuisance set: $R^2([Z_{\mathrm{nuis}}, \mathrm{LM}]) - R^2([Z_{\mathrm{nuis}}])$. The semipartial R^2, not the partial. The only predictive score this work reports.

Trained−untrained gap

The primary statistic: the unique R^2 of a language-trained model minus that of a same-architecture untrained network, under identical nuisance and splits. Isolates what *language training* adds over random initialisation.

Z_{nuis} (nuisance set)

The low-level nuisance variables subtracted in both arms of the gap: utterance length, position, word and phoneme rate, word duration, log word-frequency, and a static word-embedding block (eng1000). Surprisal is deliberately excluded (it is itself LM-derived).

Noise ceiling

The maximum variance any model could explain, set by the reliability of the neural measurement itself; reported here as split-half reliability of a held-out repeated story.

Permuted twin

The per-kind shuffled-target null that tests brain-*specificity*. It belongs to the lever question (whether the signal can be *moved*), not to the reality question, and so does not appear in the Results section written here.

$\mathbb{E}[Y \mid S]$

The stimulus-predictable mean response: the strong conditioning variable behind the ceiling question, as opposed to the weak low-level proxy Z_{nuis}.

$\mathcal{L}_{\text{brain}}$
The brain-alignment training loss: the term added to a language objective that rewards representations predictive of recorded brain activity.

Knowledge distillation (KD)
Training a small student model to match a larger teacher's output distribution rather than the raw text labels. Pure logit KD is the perplexity-only form (temperature-scaled KL on the teacher's next-token logits, no hidden-state or brain term).

Cold-init / warm-init KD
A *cold-init* student is randomly initialised before KD, so any alignment it ends with was transmitted through the distillation channel; a *warm-init* student starts from a pretrained checkpoint and is therefore a fine-tuning-drift control, not a test of what KD carries.

Headroom (Δ)
The teacher-minus-student gap in unique R^2, $\Delta = A_T - A_S$, with A_T the teacher's alignment and A_S the student's. The alignment a perplexity-only objective leaves below the teacher.

Floor-anchored retention (ρ')
The fraction of the teacher's alignment a student keeps above the untrained floor, $\rho' = (A_S - A_0)/(A_T - A_0)$, where A_0 is the untrained score: $\rho' = 0$ is the floor (destroyed), $\rho' = 1$ is the teacher (preserved). Distinct from the partial correlation $\rho_{XY \cdot Z}$ of the methods (the primed symbol is retention, never a correlation).

Data-processing inequality (DPI)
For a Markov chain $A \to B \to C$, no processing of B can increase information about A: $I(A; C) \leq I(A; B)$. Here it bounds a fixed-function compression to lose, never create, alignment relative to its teacher.

1 Introduction

Brain alignment has become a way to study what language models and the human language system share [1–3]. Linear encoding studies ask how well a model's internal activations predict human brain responses to the same text, and the resulting score has been used to compare architectures, layers, training objectives, scale, and compression [2, 4–7]. Recent work also shows that the score is fragile: it can reflect linguistic structure, but it can also be inflated by low-level stimulus features, shallow acoustic cues, or weak evaluation splits [8–10]. Together, this literature has made brain alignment useful as a measurement. This thesis asks whether the same signal can be useful as a training signal. In particular, we study model compression: during knowledge distillation, can a small student be guided by a brain-alignment objective, and does that guidance preserve anything ordinary text-only compression loses?

Problem. We focus on model compression. Knowledge distillation trains a small *student* model to imitate a larger *teacher*, reshaping the student's internal representations while it learns. That is where the alignment question becomes sharp: if brain alignment lives in middle-layer structure, can a brain-alignment objective shape the student while it is being distilled [2, 7]? We ask whether such a student preserves or improves alignment at a matched compute budget, and whether any gain matters beyond the alignment score itself. This case matters because compression is how small, deployable models are built, and recent brain-tuning and compression work has left this training use of brain alignment open [7, 11–13].

Hypothesis. The hypothesis is that brain alignment can act, at most, as a weak prior over text-consistent representations. The reason is a severe mismatch in scale. A language model learns from on the order of a trillion tokens of text. A language-fMRI dataset offers a few thousand noisy recordings, and a low noise ceiling caps the variance any model can explain in

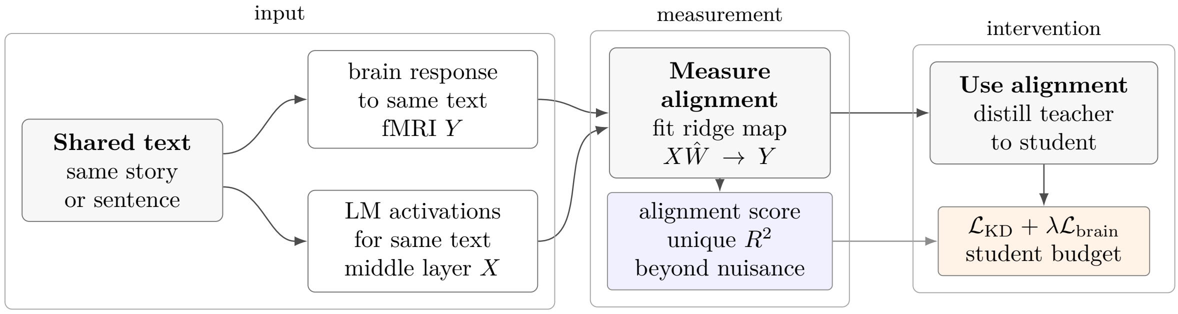

Figure 1: The measurement instrument and its inversion. A ridge encoding model maps language-model activations to brain responses; held-out accuracy is reported as unique variance over the nuisance floor. The thesis asks whether a corresponding brain-alignment term can guide a distilled student.

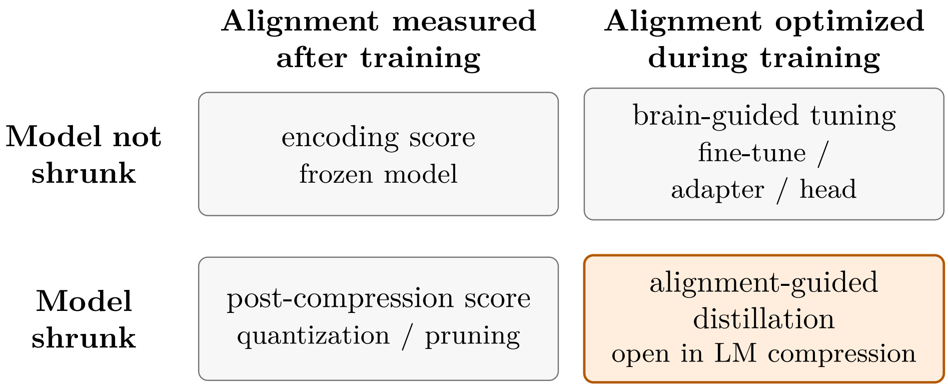

Figure 2: The field on two axes; three cells are occupied and the fourth, where this work sits, is open in language-model compression. Examples and citations are given in Table 1.

them [1, 14, 15]. Against that imbalance, the brain term is too small to carry new task structure by itself. Its plausible role is to bias the student among solutions already compatible with the text objective, the way a weak regularizer does [16–19]. This gives a falsifiable prediction. A brain prior should help most where the text signal is weakest, in small, heavily compressed, or data-starved models, and should wash out wherever text data is abundant. Section 3 makes the argument precise.

Recent studies have already used fMRI as an optimization signal for speech and text models [11, 12, 20, 21]. The case is strengthened by causal evidence: text models pushed to be brain-*misaligned* at matched perplexity perform worse across a broad downstream suite [13]. Those studies live on the grow-and-optimize side of the field: they fine-tune the model, add adapters, or attach a brain-prediction head. Compression work has taken the complementary path. It shrinks a language model first, through quantization or pruning, and measures the alignment score afterward [7]. Alignment-guided distillation combines the two: the model is shrunk, and the alignment score enters the training objective. That is the open case in Figure 2, and it is exactly the regime where the weak-prior hypothesis says a brain signal has its best chance to matter.

Brain-alignment scores are easy to inflate. Shuffled splits leak through the temporal autocorrelation of fMRI, and low-level features such as sentence length carry much of an apparent score, so an untrained network can appear to explain brain data on its own [9, 10]. We therefore hold every number in this thesis to a fixed standard, set before measuring anything. The contributions below summarize that standard, and Section 3 defines it [2, 3, 8–10].

Contributions.

This thesis makes four contributions.

- It formulates alignment-guided distillation and an information-budget prediction for when a weak neural prior could matter.
- It establishes an artifact-resistant measurement protocol and a conditional within-subject result that survives whole-story splits, a capacity-matched nuisance floor, and an untrained-network control [E006].
- It shows that an averaged-target gain does not generalize to a per-individual effect, and tests that conclusion across adapter-capacity, loss-form, and voxelwise-mechanism variants [E008][E011][E013].
- It bounds practical payoff and synthetic-proxy transfer: no OOD-perplexity benefit is detected when the alignment manipulation is unreliable, a gaze-privileged route fails its preconditions, and a learnable dense neural proxy does not improve the common real-brain endpoint [E009][E016][E024].

The program asks five questions in sequence, each only if the previous one holds. Is the signal real, or an artifact? Does plain distillation preserve it? Does an alignment objective move it beyond what perplexity already implies? Does any gain survive per individual at matched perplexity? And does it improve a downstream task?

The resulting arc is conditional rather than universal. The signal survives the recorded controls within UTS03 [E006]; plain distillation leaves alignment headroom but is quality-entangled [E003]; the tested objective does not show a reliable fold-level gain [E004]; its averaged-target positive does not generalize across participants [E008]; and neither the tested practical routes nor the dense synthetic proxy establishes real-brain benefit [E009][E016][E024].

2 Related work

This thesis sits in the one empty cell of a two-by-two grid: grow or shrink the model, and measure or optimize alignment (Table 1). We review the three occupied cells in turn, then the measurement standard this work adopts.

Table 1: The field on two axes: what an intervention does to the parameter budget, and how it treats brain alignment. Three cells are occupied; the fourth, shrinking a model while alignment is in the objective, is empty in the 2026 literature and is where this work sits.

	Alignment measured	Alignment optimized
Model grown	classical brain-score [2, 5, 6]	brain-tuning [11–13]
Model shrunk	compression robustness [7]	**this thesis (empty in the literature)**

2.1 The classical brain-score literature

The largest body of work fits encoding models to frozen, pretrained networks, then reads off how alignment varies with architecture, scale, and training. Three findings recur. Alignment lives in the structured middle layers, where syntax drives the layerwise trend and semantics is more region-specific [2]. It grows with model scale and saturates around a few billion parameters, and at a matched parameter count scale moves it more than instruction-tuning does, which is why a base model is the right teacher to align against [5]. Beyond that scale the trend can flatten or reverse, as the largest models outgrow the human language network and track formal competence more than behavior [6]. This line of work shows the signal exists and is structured, but it never intervenes through alignment, so it cannot say whether alignment is usable.

2.2 Brain-tuning

A second line puts alignment in the objective of a fine-tuned or grown model, across modalities. For speech, a voxelwise brain loss fine-tunes compact encoders and improves both alignment and several downstream tasks, with a multi-participant variant reporting further gains at higher data efficiency [11, 20]. For text, training GPT-2 and LLaMA-2 with a cosine brain loss beats text-only baselines, though the gains are mixed and the splits are shuffled with no brain-shuffled control, so they would not clear the measurement bar this work adopts [12]. The strongest evidence is causal. Training a text model to be *less* brain-aligned, at fixed perplexity, makes it worse across a suite of more than two hundred tasks, while training it to be *more* aligned improves performance on the same suite [13]. Alignment is therefore doing causal work, not just riding along with a good model. The bare inverted idea is not novel, but no study here compresses a model: every method grows or holds the parameter budget, leaving the compression question open.

2.3 Alignment under compression

The closest prior work to this thesis is the compression study of Oota et al. [7], which tracks how alignment changes as a model is compressed. They find that alignment survives only certain post-training quantizers on models of at least a few billion parameters; other quantizers, smaller models, and pruning all degrade it, and distillation is never tested. That study measures alignment under compression but never optimizes it. It reports how quantization and pruning move the score, and never makes alignment a training target. Distillation is a different and lossier operation. It refits a smaller model from scratch against the teacher, the step most likely to destroy the middle-layer structure where alignment lives. So alignment surviving quantization, a gentle reshaping of fixed weights, does not imply it survives distillation. Whether distillation preserves alignment is therefore an open empirical question: how much does it lose by default, and can a brain objective recover the rest? We develop the information-theoretic bound behind this in Section 3.

2.4 The empty cell

No prior work occupies the empty cell. Two nearby precedents, one in vision and one in audio, are sometimes cited as indirect support for a brain regularizer when data is scarce, but neither isolates a brain-specific effect. Pirlot et al. [18] regularize a vision network with neural data and improve both accuracy and adversarial robustness, yet a shuffled-label control reproduces the accuracy gain, so only the robustness benefit depends on real neural structure. Freteault et al. [19] fine-tune a small audio network on fMRI and improve most downstream tasks, but without an interaction test or a matched non-brain control, the low-data advantage is not isolated. Neither study is in the distillation regime this work targets, so both inform our design without testing the claim.

2.5 The measurement bar

A separate line of work warns that alignment scores are easy to inflate and over-read, and it sets the standard this work adopts. Shuffled splits leak through the temporal autocorrelation of fMRI, low-level features like sentence length carry much of the apparent signal, and some published scores fail basic predictive and reliability checks [9, 10]. The same headline score can reflect real semantics in one model and shallow phonology in another, so the number alone does not pin down the mechanism [8]. A differentiable, fMRI-free surrogate would go further, but the literature does not agree on its direction: local intrinsic dimension correlates with alignment in opposite ways across language and vision, and similarity-based surrogates track width and depth as much as alignment [22–24]. We therefore leave a surrogate to future work, since real fMRI must first establish which way to push it.

3 Methods

3.1 The alignment score

We measure alignment with a ridge encoding model that maps a frozen model's per-stimulus middle-layer activations X to recorded brain activity Y, fit on training stimuli and scored on held-out ones. For activations $X \in \mathbb{R}^{n\times d}$ over n stimuli and a v-voxel response Y, the fitted map is

$$\hat{W} = \arg\min_{W} \|Y - XW\|_F^2 + \lambda\|W\|_F^2, \tag{1}$$

and the score for each voxel is the correlation between its predicted and measured response. A raw score of this kind means little on its own, because a model can predict the brain for reasons that have nothing to do with language: utterance length, word position, generic word identity. We therefore report not the raw score but the variance the model adds *on top of* a low-level nuisance set Z_{nuis}, the difference of two cross-validated fits,

$$\Delta R^2 \;=\; R^2\big([Z_{\text{nuis}}, \text{LM}]\big) - R^2\big([Z_{\text{nuis}}]\big). \tag{2}$$

A positive ΔR^2 says the contextual representation explains brain activity that length, rate, and static word identity cannot [E002].

3.2 What the score measures

We use the score as a linear lower-bound proxy for an information increment: the information the model's representation carries about brain activity beyond the low-level baseline. The underlying estimand is conditional mutual information. For three variables (X, Y, Z),

$$I(X; Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z) \tag{3}$$

is the dependence between X and Y that remains once Z is known [17]. Conditioning is essential here. Plain mutual information cannot distinguish the dependence of interest from a shared driver: if X and Y are both driven by some common Z, then $X \perp Y \mid Z$, so the conditional term is zero even when the unconditional term is positive. The chain rule splits the dependence into a baseline and the increment over it,

$$I\big(B; [Z_{\text{nuis}}, \text{LM}]\big) = I(B; Z_{\text{nuis}}) + I(B; \text{LM} \mid Z_{\text{nuis}}), \tag{4}$$

with B the brain response, so the information the model's representation adds over the nuisance is exactly the conditional term $I(B; \text{LM} \mid Z_{\text{nuis}})$.

The conditional mutual information is not directly measurable from data, so we estimate it through the unique-variance score, which is its linear-Gaussian surrogate. Under joint Gaussianity, conditional mutual information is a monotone function of the squared partial correlation,

$$I(X; Y \mid Z) = -\tfrac{1}{2}\ln\big(1 - \rho^2_{XY\cdot Z}\big), \tag{5}$$

so the quantity is positive exactly when the partial correlation is nonzero [17, Ch. 8]. The reported score is the *semipartial* R^2, the raw increment, which differs from the partial ρ^2 only by the factor $1/(1 - R^2([Z_{\text{nuis}}]))$. Both vanish together, so a positive semipartial score is equivalent to a positive conditional mutual information. Ridge is linear, so the score captures the linearly decodable part of that information, a population-level lower bound.

The choice of Z fixes what the claim means. Conditioning on the low-level nuisance Z_{nuis} shows the signal is real beyond low-level features, not beyond the stimulus as a whole. Conditioning instead on the full stimulus-predictable response $\mathbb{E}[Y \mid S]$ asks a stronger question: not whether the representation explains brain activity beyond low-level features, but whether it explains anything the stimulus itself does not already predict. That is the stimulus-predictability ceiling, and it reuses the same chain rule with $\mathbb{E}[Y \mid S]$ in place of Z_{nuis} as the larger conditioning set.

3.3 Why a brain signal is only a weak prior

A brain term in a language objective is, up to constants, a maximum-a-posteriori estimate. Training chooses parameters θ that minimise a loss, and adding a brain term to the usual language-model loss gives

$$\hat{\theta} \;=\; \arg\min_{\theta} \; \underbrace{\Big[-\textstyle\sum_{t=1}^{N} \log p_\theta(x_t \mid x_{<t}) \Big]}_{-\log p(\mathcal{D}_{\text{text}}|\theta)} \;+\; \lambda \, \underbrace{\mathcal{L}_{\text{brain}}(\theta)}_{-\log p(\theta)} . \tag{6}$$

The first term is the negative log-likelihood of the text: next-token prediction is maximum likelihood, and minimising it maximises $p(\mathcal{D}_{\text{text}} \mid \theta)$. The second term is the negative log-prior: any regulariser that penalises a property of θ corresponds, up to a normalisation constant, to a prior $p(\theta) \propto \exp\big(-\lambda \mathcal{L}_{\text{brain}}(\theta)\big)$ over parameters that favour that property, with λ setting its strength. Minimising their sum is therefore

$$\hat{\theta}_{\text{MAP}} \;=\; \arg\max_{\theta} \; \big[\log p(\mathcal{D}_{\text{text}} \mid \theta) + \log p(\theta) \big], \tag{7}$$

a MAP estimate in which the text is the likelihood and the brain term is the log-prior. But how far can a prior that small move a posterior dominated by the text?

A training corpus carries on the order of 10^{12} to 10^{13} bits of supervision. The brain side offers a few thousand noisy stimuli, and the information each voxel can carry is capped by its noise ceiling, a channel capacity

$$I \le \tfrac{1}{2} \log_2(1 + \text{SNR}), \qquad \text{SNR} = \frac{r_{\max}^2}{1 - r_{\max}^2}, \tag{8}$$

which puts the usable brain budget, as an order of magnitude, around 10^4 to 10^6 bits [17]. The data-processing inequality shrinks it further. Let S be the stimulus and B its noisy brain response. Under the modelling assumption that B is generated from S, the chain $\theta^\star \to S \to B$ gives

$$I(\theta^\star; B) \le I(\theta^\star; S). \tag{9}$$

The brain response cannot contain more information about the target parameters than the stimulus that elicited it. Such a small prior cannot teach new task structure the text does not already contain. It can only select among the many near-equivalent solutions an overparameterized model admits, which defines the role of the brain term as a regularizer rather than a source of new information.

A generalization bound can show quantitatively how and where such a weak prior can help. Modelling training as a channel from the data Z^n to the learned parameters θ, the expected generalization gap is bounded by the mutual information between them,

$$\overline{\text{gen}} \;\le\; \sqrt{\frac{2\sigma^2 \, I(\theta; Z^n)}{n}}, \tag{10}$$

for σ-sub-Gaussian loss [16]. If the brain prior adds ΔI bits to $I(\theta; Z^n)$, its benefit to generalization is at most $O(\sqrt{\Delta I / n})$, and it shrinks as the task data n grows. A weak but well-targeted brain prior should therefore help most where the text signal is weakest, in small, under-trained, or aggressively compressed models, and fade where text data already dominates. Distillation is the regime where that prediction is most testable.

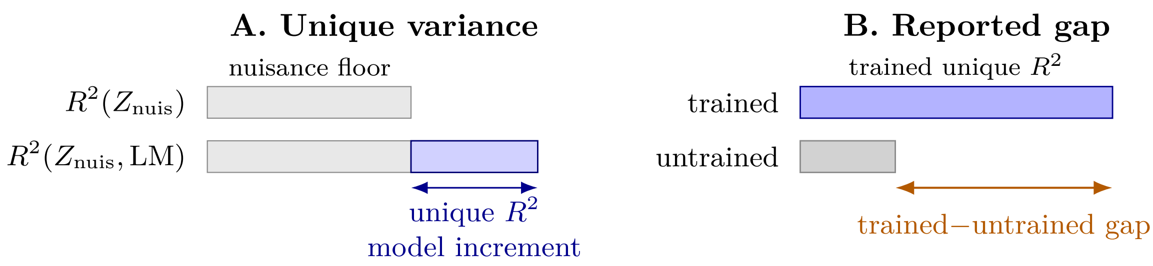

Figure 3: Schematic of the two computed quantities. Unique variance is the held-out R^2 increment from adding LM features to the nuisance model. The reported gap is $\Delta R^2_{\text{trained}} - \Delta R^2_{\text{untrained}}$, with both arms scored using the same nuisance set and splits. Bar lengths are illustrative.

3.4 The trained–untrained gap

We report the *gap* between a trained model and a same-architecture untrained network, both fit with the same nuisance set and the same splits, rather than the trained model's absolute score. An absolute score is insufficient on naturalistic data, because even an untrained network can score slightly positive. The stimulus drives both the brain response and the untrained features; if the nuisance set captures only part of the stimulus, residual shared-cause dependence can remain. Subtracting the untrained arm subtracts an estimate of the component shared by architecture, tokenisation, and residual temporal structure. The gap therefore estimates what *language training* adds over random initialisation [E006].

3.5 The controls

We apply three controls uniformly. *(1) Contiguous splits*: train and test are divided in blocks (block cross-validation on the ROI benchmark, whole-story-held-out cross-validation on the voxelwise run), because shuffling time-adjacent fMRI samples leaks near-duplicate timepoints across the split and inflates the score [9][E002][E006]. *(2) A capacity-fair nuisance floor*: the wide contextual block and the wide static-embedding block are each reduced by PCA to the same rank, fit on the training fold only, so the language model does not win merely by carrying more dimensions; the few low-level scalars stay raw [9, 10][E006]. *(3) A held-out noise ceiling*: on the voxelwise run only reliable voxels are kept, selected by split-half reliability on a separate story repeated ten times and held entirely out of the cross-validation pool, so the estimate is not inflated by voxels that happen to correlate with the model by chance [14, 15][E006]. Surprisal is deliberately left out of the nuisance set, since it is itself LM-derived and subtracting it would remove part of the construct under test; log word-frequency, a static lexical scalar, is distinct and is subtracted [E006]. The block-permuted-target twin reorders contiguous response blocks and enters only where the question is whether the signal can be moved [11, 18]. It is a sensitivity control rather than a complete randomization null: the implementation can leave some rows paired with their original targets, so no conclusion relies on calling it fully permuted or exactly marginal-preserving [E016].

3.6 Data and models

We use two datasets. The Tuckute ROI benchmark uses the 1000 isolated sentences and five left-hemisphere language regions of Tuckute et al. [1]. It averages responses across training participants, so it is coarse, but its isolated-sentence design avoids the temporal-autocorrelation leakage of naturalistic time series; its reported noise ceiling is $r \approx 0.35$ [E002]. The voxelwise measurement uses subject UTS03 of the deep-sampled naturalistic story-listening dataset of LeBel et al. [14]: 20 stories, story-level folds, and 11,442 voxels retained at split-half reliability above 0.5 on a held-out repeated story [E006]. Its scientific scope is conditional on this participant and protocol: spatially dependent voxels increase measurement resolution but are not independent replicates, and the five folds share training stories [E006]. The models are GPT-2, GPT-2-medium, and Qwen2.5-0.5B, each read at its alignment-peak middle layer, with a same-architecture untrained network as the control in every comparison [E002][E006].

3.7 Distillation and the brain term

The compression setting is knowledge distillation: a smaller *student* learns to reproduce a larger *teacher*'s next-token distribution, rather than learning from text directly. Perplexity gives the output-quality control. Per-token cross-entropy is code length, maximum-likelihood training minimizes the KL divergence from the data distribution to the model's, and perplexity is the exponential of that cross-entropy [17]. Both objectives are KL objectives with different references: plain language-model training projects the student onto the empirical data, while distillation projects it onto the teacher's softened distribution. Matching *held-out perplexity* across arms holds output quality fixed while the objective varies; matching compute does not.

The distillation objective does not by itself imply preserved alignment, for two reasons. First, it constrains the *output* distribution, whereas alignment is a property of the *internal* representation, and the readout from representation to logits is many-to-one,

$$D\big(p_T \,\|\, q_S\big) \to 0 \;\not\Rightarrow\; Z_S \approx Y_{\text{teacher}} \;\not\Rightarrow\; I(Z_S; B \mid Z_{\text{nuis}}) = I(Y_{\text{teacher}}; B \mid Z_{\text{nuis}}), \tag{11}$$

so matching the teacher's output need not match its representation. Second, a from-scratch student also fits the stimulus through the raw next-token targets, a route the teacher's features do not mediate. That route breaks the Markov chain behind the data-processing bound, so the teacher's alignment no longer upper-bounds the student's. Whether distillation preserves alignment is therefore an empirical question, not a corollary.

The brain term enters the objective as a co-trained regression loss,

$$\mathcal{L} = \mathcal{L}_{\text{KD}} + \lambda\, \mathcal{L}_{\text{brain}}, \tag{12}$$

with mixing weight λ. The primary test arm is a randomly initialised student, because any alignment it retains must be learned during distillation rather than inherited from pretraining. A warm-initialised student and a no-teacher fine-tune are kept as labelled drift controls, and every arm is scored at one fixed layer within a single architecture so no layer choice biases the comparison [E003]. Brain *specificity* is tested with a permuted-target twin: the same loss, but against brain targets shuffled across stimuli. The arms share corpus, batch size, learning rate, and architecture, but they are not fully compute matched: the from-scratch arm runs about twice the optimizer steps of the warm arms, a caveat carried into the results [E003].

3.8 Intervention tests and inference units

Q2 and Q3 compare a real-target brain-loss arm with its block-permuted twin, always pairing arms within the same stimulus fold and stochastic seed. For Q3, the model is trained separately against each participant's response rather than against a response averaged across participants. The participant-level estimand is the real-minus-permuted change in held-out unique R^2, aggregated within participant over seeds and five shared stimulus folds; inference is reported across nine participants with a separate fold-clustered sensitivity analysis [E008]. The effect on a response averaged across participants is a different estimand because unique R^2 does not commute with response averaging [E005][E008].

Higher-rank LoRA and a contrastive loss test capacity and objective robustness at the same ROI-scale operating point [E011][E013]. The naturalistic voxelwise experiment instead asks whether the MSE distillation manipulation takes hold at all across a fixed weight grid. Its neutral-to-harmful trajectory is a lever failure, not a participant-level null or evidence against every possible voxelwise intervention [E013].

Q4 first tests whether a reliable brain-specific representation change exists and then measures downstream change as an out-of-domain perplexity contrast across eight training seeds [E009]. A separate privileged-information route is gated before model training: the gaze signal must predict the label without task-induced leakage, and the corpus must support a stable paragraph-disjoint learning-curve split [E024].

3.9 Synthetic-target training and real-brain transfer

E016 tests whether a dense brain-derived synthetic target changes the data constraint. GPT-2-medium is distilled into GPT-2 on 95,999 WikiText training sentences, with 1,999 held out, six stochastic seeds, and target weight $\lambda = 0.1$ [E016]. The TRIBE target is a 20,484-dimensional predicted-cortical-response vector read at student layer 6. The comparison target, textfeat, projects a frozen teacher hidden state to the same nominal output width. It is dimension-matched but not demonstrated matched-information: its effective rank is at most 1024, its geometry differs from TRIBE, and the KD baseline fits the two endpoints at substantially different levels [E016]. Held-out target R^2 is therefore interpreted within each target family rather than as a common cross-family scale.

Saved students then pass through a separate real-brain transfer gate on the Tuckute condition-B responses: 1000 sentences, a fixed response averaged over five participants and five left-hemisphere language ROIs, and five contiguous nuisance-subtracted folds [E016]. The transfer readout uses student layer 7, explicitly different from the layer-6 synthetic training target. For each seed, the primary contrast is the Tuckute gain of TRIBE training over KD minus the corresponding gain of textfeat training over KD. The inference unit is six stochastic training seeds over one fixed participant-averaged endpoint, so the exact sign test supports no participant- or population-level inference. Corresponding KD weights are byte-identical across target jobs, PCA and scaling are fit on training data, and normalized WikiText-train, WikiText-heldout, and Tuckute sentence sets have zero exact overlap [E016].

4 Results

4.1 The alignment signal is real beyond low-level artifacts

A trained middle layer predicts held-out language-network activity beyond low-level stimulus features, while a same-architecture untrained network does not. The reality check matters because the later intervention questions only make sense if the measured score is not a low-level artifact. The Tuckute ROI benchmark shows the effect across three language models, and the LeBel UTS03 run repeats it at voxel scale within one deeply sampled participant.

The Tuckute ROI benchmark uses the five left-hemisphere language regions of Tuckute et al. [1], 1000 isolated sentences, and five contiguous folds. The trained-minus-untrained gap is positive for all three models, while the untrained control is negative at every layer and seed (Table 2). The positive signal spans a broad plateau of middle layers and disappears for random features [E002].

Table 2: Tuckute ROI benchmark (five LH language ROIs, five contiguous folds). Unique R^2 is the semipartial increment over the nuisance set; the gap is trained minus untrained. NC-norm is the gap as a fraction of the $r \approx 0.353$ noise ceiling.

Model	Best layer	Trained ΔR^2 (mean $\pm$ SD)	Untrained ΔR^2	Gap	NC-norm	Source
GPT-2 (12L)	L7	$+0.020 \pm 0.008$	-0.011	+0.030	0.056	[E002]
GPT-2-medium (24L)	L14	$+0.019 \pm 0.005$	-0.014	+0.033	0.053	[E002]
Qwen2.5-0.5B (24L)	L12	$+0.036 \pm 0.010$	-0.014	+0.050	0.102	[E002]

The voxelwise run uses subject UTS03 of LeBel et al. [14], 20 naturalistic stories, and 11,442 voxels selected at split-half reliability above 0.5 on a held-out repeated story. The main readout is the trained-minus-untrained gap: +0.0207 for GPT-2 and +0.0277 for Qwen2.5-0.5B (Table 3). The sign is also widespread across voxels: 95% of reliable GPT-2 voxels and 99% of reliable Qwen voxels have positive gaps. The untrained unique R^2 is -0.017 in both comparisons [E006]. For both models the gap is positive on all five held-out story blocks. The folds share training stories, and the voxels are spatially dependent measurements from one participant, so these signs are sensitivity evidence rather than independent population replicates [E006].

Table 3: Voxelwise run on LeBel UTS03 (20 stories, story-level folds). The gap is the trained-minus-untrained unique R^2. The percentage column describes spatial prevalence within UTS03; the final column reports fold-block sensitivity, not independent participant replication.

Model	Layer	Trained – untrained gap	% reliable voxels gap > 0	Positive story blocks	Source
GPT-2	L7	+0.0207	95%	5/5	[E006]
Qwen2.5-0.5B	L12	+0.0277	99%	5/5	[E006]

The controls address different ways a low-level artifact could mimic alignment. The Tuckute ROI benchmark uses isolated sentences, which removes temporal autocorrelation from naturalistic time series and makes a positive result harder to obtain [E002]. The voxelwise run uses naturalistic stimuli but holds out whole stories; test timepoints are never adjacent to training timepoints. Voxel selection uses a separate repeated story, preventing the analysis from selecting voxels because they happen to correlate with a model on the evaluation stories [E006]. Both trained and untrained scores subtract word and phoneme rate, word duration and length, log word frequency, and a static word-embedding block, the low-level feature set most likely to inflate the score [E006]. Surprisal and imageability are deliberately left unsubtracted; the nuisance baseline therefore does not remove every contextual property of the stimulus. The same-architecture untrained network supplies the stronger check: on the same stimuli and nuisance baseline, it scores negative at -0.017 [E006]. Architecture, rate, and residual temporal structure therefore cannot explain the positive trained-minus-untrained gap alone [E006].

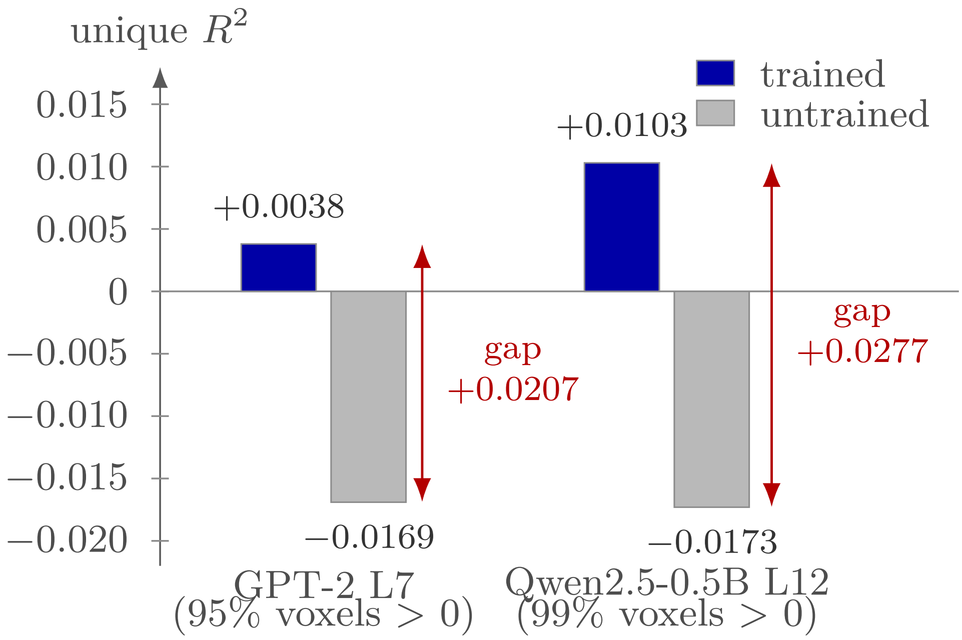

Figure 4: Mean unique-R^2 on LeBel UTS03 reliable voxels. Trained GPT-2 L7 and Qwen2.5-0.5B L12 exceed same-architecture untrained controls; arrows show trained-minus-untrained gaps, and group labels report the fraction of reliable voxels with positive gaps [E006].

Together, the Tuckute ROI benchmark and the voxelwise UTS03 run show that the alignment score contains a language-training signal under the recorded controls. The Tuckute benchmark supplies breadth across models, while UTS03 supplies spatially resolved within-subject evidence. The original voxel-bootstrap intervals are retracted because they treated spatially dependent voxels as independent units; cross-subject generality remains open [E006]. Qwen aligns more strongly than either GPT-2 variant, while the two GPT-2 variants are nearly tied despite a threefold parameter difference. The single cross-family contrast alone cannot establish a quality ordering; Section 4.2 takes up the quality question directly [E002]. Q0 therefore gives the rest of the program a valid measurement target. The result does not show that the target can be moved during training or that moving it improves a model.

4.2 Plain distillation does not preserve alignment by default

Knowledge distillation (KD) trains a smaller student to match a larger teacher's output distribution. E003 asks whether that ordinary perplexity-only objective keeps the teacher's brain alignment. The from-scratch logit-KD student lands far below its teacher, with a gap of +0.0181 ($p < 0.001$), and retains only 0.37 [0.14, 0.58] of the teacher's alignment above an untrained floor [E003]. The cold-init drop rules out automatic preservation and gives the later alignment-guided objective real headroom. That same E003 comparison also limits the claim: among the five non-teacher arms, alignment tracks log-perplexity (Pearson −0.88), so the retention gradient could reflect language-model quality rather than the distillation objective itself [E003][L011].

The headroom is $\Delta = A_T - A_S$, where A_T is the teacher's alignment and A_S is the student's alignment. Both quantities use the same measurement protocol as Q0: unique R^2 on the Tuckute ROI benchmark, contiguous folds, and the capacity-fair nuisance baseline. KD has to be measured directly because it follows a different path from fixed-function compression. For deterministic post-processing of a fixed teacher, such as the quantization and pruning studied by Oota et al. [7], the student's features depend on the stimulus through the teacher's features. The data-processing inequality then bounds $I(Z_{\text{student}}; B) \le I(Y_{\text{teacher}}; B)$ along that teacher-to-student path. A from-scratch KD student has another route: gradient descent on the teacher's

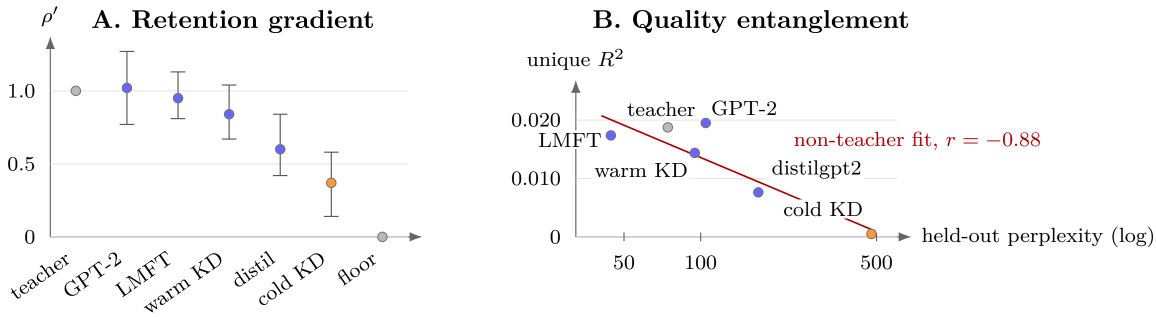

Figure 5: Plain KD leaves alignment headroom, with the loss entangled with model quality. (A) Floor-anchored retention ρ' falls along the rebuild ordering, from teacher-level pretrained references toward the untrained floor; the distilgpt2 retention follows the E003 lineage-teacher denominator. Error bars are bootstrap intervals where recorded. (B) Unique R^2 co-varies with held-out perplexity; the teacher is plotted for context, and the line is the recorded fit over the five non-teacher arms with Pearson coefficient -0.88 [E003].

logits still exposes the student to the training text. The fixed-function premise no longer holds, so preservation becomes an empirical question [E003].

The cold-init student is the key test. A warm-started student already inherits teacher-level alignment from pretraining, which Q0 measured directly. High retention after a few KD epochs would mainly measure fine-tuning drift from that starting point [E003]. The cold-init GPT-2 begins from random weights and is trained on the teacher's logits, so any surviving alignment must pass through the distillation run. The warm-init KD arm and the no-teacher plain-LM finetune remain as drift controls. All comparison arms use the 12-layer GPT-2 architecture and one fixed layer, so cross-architecture layer choice cannot bias the comparison [E003].

Table 4: Knowledge-distillation arms on the Tuckute ROI benchmark at the fixed alignment layer [E003]. The ΔR^2 column reports raw unique R^2; E003 also reports NC-normalised values using noise ceiling 0.353. Floor-anchored retention $\rho' = (A_S - A_0)/(A_T - A_0)$ is 0 at the untrained floor and 1 at the teacher; for distilgpt2, ρ' and Δ use GPT-2, its lineage teacher. p(no-drop) is the bootstrap probability of no drop from the teacher.

Arm	Init / objective	ΔR^2	ρ' [boot. CI]	Δ vs teacher	p(no-drop)	ppl	Source
teacher gpt2-medium	pretrained	+0.0188	n/a	n/a	n/a	74	[E003]
gpt2	pretrained (conventional)	+0.0195	1.02 [0.77, 1.27]	≈ 0	0.57	105	[E003]
lmft_warm	warm + plain-LM finetune	+0.0174	0.95 [0.81, 1.13]	+0.0016	0.265	44	[E003]
kd_warm	warm + pure logit KD	+0.0144	0.84 [0.67, 1.04]	+0.0046	0.061	95	[E003]
distilgpt2	real distillation (+hidden-cosine)	+0.0076	0.60 [0.42, 0.84]	+0.0119	0.001	169	[E003]
kd_cold	**random + pure logit KD**	+0.0005	0.37 [0.14, 0.58]	+0.0181	0.000	477	[E003]
untrained	random (floor)	−0.0102	0	n/a	n/a	$\sim 6 \times 10^4$	[E003]

The supported finding is the monotone retention gradient (Table 4). Conventional GPT-2 sits at the teacher's level ($\rho' \approx 1.0$), warm-init KD retains 0.84, the published distilgpt2 retains 0.60 relative to GPT-2, the cold-init student retains 0.37, and the untrained floor is 0 [E003]. The reference arms therefore anchor both ends of the scale: off-the-shelf GPT-2 marks teacher-level alignment, and the random model marks the floor. Two runs scoring the same references on different GPUs reproduced the reference scores to about 0.0003 [E003]. The ordering holds at every PCA rank tested ($n_{\text{pca}} \in \{25, 50, 100\}$). The absolute magnitudes are less stable: the two quality-entangled arms change sign in unique R^2 across ranks, so the ordering is the durable finding and the loss fractions are rank-dependent [E003].

The quality caveat is important. Alignment co-varies with log-perplexity across the same non-teacher arms (Pearson -0.88). In that fit, warm-init KD lies essentially on the regression line, while conventional GPT-2 is the only arm more aligned than its perplexity predicts.

Two contrasts would separate objective-specific loss from a quality explanation, and both are marginal. Warm-init KD is less aligned than GPT-2 despite lower perplexity, with bootstrap $P(\text{gpt2} \leq \text{kd\_warm}) = 0.092$ [E003]. Among the warm-start controls, the no-teacher finetune beats warm-init KD on both perplexity and alignment, with $P(\text{lmft} \leq \text{kd\_warm}) = 0.085$ [E003]. Both contrasts lean on a single unreplicated point and, under a paired-fold test, on one fold carrying about 44% of the signal [E003][L011]. The cold-init arm also remains under-trained: its held-out perplexity is 477, far above the converged arms' 44 to 95. Part of its near-floor alignment may therefore reflect a weaker language model rather than the distillation objective itself. E003 predeclared that convergence guard [E003]. The arms were matched on corpus, batch size, learning rate, and architecture, but the from-scratch arm ran about twice the optimizer steps of the warm arms. The step imbalance deepens the under-training caveat instead of removing it [E003].

Plain perplexity-only distillation therefore leaves genuine headroom below the teacher. E003 does not decide whether an alignment-guided objective can recover that headroom beyond what perplexity already predicts. The evaluation is an ROI benchmark (NC ≈ 0.35), not a matched-perplexity or fully optimizer-step-matched causal test. The Tuckute ROI benchmark is also underpowered for a ~ 0.005 objective-specific gap. The matched-perplexity question therefore requires a paired design whose inference unit matches the intended claim [E003][L011].

4.3 An alignment-guided objective has not yet shown a reliable held-out gain

E004 tests whether a co-trained MSE brain term changes held-out alignment beyond an otherwise identical block-permuted-target twin. The primary Qwen arm has a mean real-minus-permuted contrast of +0.00316, but its five-fold interval is [−0.00227, +0.00858] and includes zero [E004]. The earlier positive interval treated three seeds by five folds as 15 independent observations even though seeds reuse the same fold data. Averaging seeds within each fold leaves five units; fold 4 contributes 64% of the summed contrast, and removing it reduces the mean to +0.0014 [E004].

No tuned arm improves reliably over the untuned base. The Qwen MSE arm changes held-out unique R^2 by +0.0025 relative to base with interval $[-0.0011, +0.0067]$, while all GPT-2 brain-loss arms are non-positive relative to base [E004]. The recorded unpaired permutation-null check also fails. Perplexity roughly doubles under LoRA tuning, so even the small positive paired contrast is entangled with language-model degradation rather than isolated at matched quality [E004].

The supported Q2 conclusion is therefore narrow: this experiment demonstrates no reliable held-out alignment lever at its valid fold-level inference unit. It does not prove that the true effect is zero, because five folds cannot exclude a small effect near the observed mean. The next question must use independent participants and matched perplexity rather than treating more seeds or voxels as biological replication.

4.4 No per-individual gain at matched perplexity, and the averaging artifact

The group-averaged target in E005 yields an apparent real-minus-permuted gain of +0.0081. E008 repeats the same comparison separately for nine participants and centers near zero: mean +0.00010, subject-axis interval [−0.00037, +0.00058], with 5/9 participant effects positive [E005][E008]. The shared-fold sensitivity is also centered near zero, with mean +0.00026 and interval [−0.0004, +0.0009]; every leave-one-fold-out check fails to establish a positive effect [E008]. The four participants whose responses contributed to the earlier averaged target have mean −0.00010 when evaluated individually, while five held-out participants have mean +0.00026 with an interval including zero [E008].

This difference is not a claim that averaging simply recovers the average participant. Averaging responses changes the target toward the shared stimulus-evoked component and raises its signal-to-noise ratio; unique R^2 is nonlinear in that operation. The valid conclusion is that the positive result belongs to the multi-participant-averaged estimand and does not generalize to a mean per-individual effect in this cohort [E005][E008].

Two variants do not change that conclusion at ROI scale. Increasing LoRA rank and epochs gives mean +0.00043, interval [−0.0005, +0.0013]; one participant supplies 79% of the summed effect, and leave-one-participant-out returns the estimate to the E008 scale [E011]. This is robustness to extra adapter capacity at the same effective operating point, not a stronger manipulation: absolute alignment and perplexity move no more than in E008. Replacing MSE with a contrastive objective gives mean −0.00019, interval [−0.0008, +0.0005], with 4/9 participant effects positive [E013].

The voxelwise naturalistic route fails earlier in the causal chain. Across its tested loss weights, real-target distillation on UTS03 is neutral at low weight and increasingly harmful at higher weight; it never improves over the untuned base or its block-permuted twin [E013]. Because the manipulation does not create an improving representation, this result is a failure of this distillation mechanism rather than a clean same-substrate population null.

The Q3 result is therefore a cross-participant null in E008, robust to the tested LoRA-capacity and ROI-loss variants, plus a separate voxelwise distillation-lever failure. It does not establish that every neural objective, dataset, or optimization regime must fail.

4.5 No detected practical payoff in the tested regimes

E009 asks whether alignment-guided KD improves out-of-domain perplexity on Pereira and LeBel text. The prerequisite representation change is itself unreliable: across eight seeds the real-minus-permuted alignment contrast has mean +0.011, median +0.0042, and 5/8 positive signs, while its measured 80%-power minimum detectable effect is 0.023 [E009]. Seed 0 contributes +0.063; the other seven seeds average approximately +0.004 [E009]. All downstream contrasts lie inside their measured minimum detectable effects, including comparisons with KD-only, the block-permuted arm, and the text-feature target [E009]. This supports no detected brain-specific OOD-perplexity payoff under the all-data, averaged-target setup. It is not a universal downstream null because the brain-specific manipulation that should mediate such a payoff never became reliable.

E024 tests a different route, train-only gaze as privileged information, but stops at its binding preconditions rather than executing the planned intervention. ZuCo natural-reading gaze has split-half reliability 0.71, yet its rich gaze features predict five-way relation type at balanced accuracy 0.202 against 0.200 chance, permutation $p = 0.35$ [E024]. Under task-directed reading the same probe reaches 0.295 against 0.100 chance, $p = 0.003$, showing that apparent informativeness can enter through task instructions rather than a clean train-only signal [E024]. The 300 natural-reading sentences also come from only seven paragraphs, so paragraph-disjoint test size varies too sharply to define a stable low-data learning curve [E024]. The gate therefore stops the five-arm intervention before compute. Q4 remains bounded to two observations: no detected OOD-perplexity benefit when the alignment manipulation is unreliable, and no valid privileged-information experiment on the gated gaze substrate.

4.6 Synthetic target fit does not transfer to real-brain alignment

On their respective held-out synthetic endpoints, TRIBE-target training improves target R^2 over KD by +0.078298, while textfeat-target training improves its own target R^2 by +0.000903 [E016]. The maximum relative perplexity deviation across the checked target and control arms is below 0.009221 [E016]. These results establish that the dense TRIBE target can be optimized under nearly unchanged language quality. They do not establish brain specificity or generic superiority over non-brain information because the two target families differ in geometry, rank, and baseline difficulty.

The common real-brain endpoint reverses the direction needed for a positive claim. The TRIBE-minus-textfeat Tuckute gain relative to KD is -0.001213; all six seed margins are negative, giving exact sign $p = 0.03125$ [E016]. The raw TRIBE-minus-KD Tuckute gain is also negative at -0.000919 [E016]. The synthetic-target improvement therefore does not transfer to improved Tuckute alignment in this six-seed diagnostic.

This conclusion is deliberately narrower than participant-level biological evidence. The endpoint is one fixed average over five participants and five ROIs, synthetic training reads layer 6 while transfer scoring reads layer 7, and textfeat is only a dimension-matched projected-teacher control. The imperfect block-permuted comparisons remain non-load-bearing sensitivity checks.

5 Discussion

The experiments separate measurability from controllability. A linear readout detects a trained-over-untrained signal under the UTS03 controls, but that fact does not imply that a small neural loss can move the representation reliably during distillation. The loss may be weak relative to the text objective, the response target may be noisy, and the optimization path may damage the language model before it creates a stable alignment change.

The inference unit changes the scientific answer. An objective-specific gain appears against a response averaged across participants, but the corresponding per-participant effects center near zero. This is not a contradiction: response averaging estimates alignment to a higher-SNR shared stimulus-evoked target, whereas the participant analysis estimates the mean individual effect. The protocol must name which of these estimands it claims before training begins.

The controls repeatedly narrow apparent positives. Seed-by-fold pooling creates a narrow Q2 interval that disappears at the fold unit; participant separation removes the averaged-target Q3 gain; and the Q4 payoff test becomes non-diagnostic when the representation manipulation itself is unstable. The contribution is therefore not a new optimizer that improves a downstream score, but a controlled account of where the training claim fails and which comparisons are needed to see that failure.

E016 supplies a separate construct-validity warning. A dense brain-derived proxy can become easier for a student to predict while a common real-brain alignment endpoint does not improve. Proxy optimization and biological transfer are distinct estimands; a training paper needs both before calling a synthetic neural target brain-specific. Here the result supports a transfer failure for this target, model pair, and diagnostic, not a claim that synthetic neural supervision can never help.

The negative findings are bounded. They cover the tested models, datasets, objectives, and quality regimes. E013 shows that this voxelwise MSE distillation lever fails to take hold, not that every naturalistic or voxelwise neural objective is impossible; E024 stops an invalid substrate before an intervention rather than reporting a completed LUPI null.

6 Limitations

The UTS03 result is conditional on one deeply sampled participant. Its spatially dependent voxels provide resolution, not population replication, and its five story folds share training data [E006]. The nuisance set subtracts the listed rate, duration, frequency, and static-semantic variables; surprisal and imageability remain outside it, so "beyond nuisance" means beyond those variables actually controlled.

E004 has only five seed-averaged folds and uses an imperfect block-permutation sensitivity control. Its corrected interval cannot exclude a small lever near the observed mean [E004]. E008 evaluates nine participants exposed to the same 1000 sentences, so the participant and stimulus axes are not jointly large [E008]. E011 changes adapter capacity at the same effective operating point rather than inducing a stronger representation change, and E013's voxelwise result is a failed MSE-distillation manipulation rather than a substrate-wide impossibility [E011][E013].

E009 measures OOD perplexity only, with per-arm language quality reported rather than exactly equalized; because its representation fulcrum is unreliable, it does not identify the payoff of a successful manipulation [E009]. E024 is a precondition result: no five-arm privileged-information intervention was completed, and its seven-paragraph corpus cannot support the intended stable paragraph-disjoint learning curve [E024]. E016 compares target families with different geometry and baseline difficulty, and its transfer inference spans six training seeds over one fixed participant-averaged Tuckute endpoint. Its block permutation is incomplete and remains outside the load-bearing transfer conclusion [E016].

7 Conclusion

Brain alignment is measurable under the fixed UTS03 control pipeline, but the tested objectives do not demonstrate a reliable per-individual training lever or practical payoff. The averaged-target positive disappears when participants become the inference unit, capacity and loss variants do not rescue it at ROI scale, the voxelwise distillation mechanism fails to create an improving manipulation, and a learnable dense neural proxy does not transfer to the real-brain endpoint. The durable result is therefore methodological and scientific: alignment-guided training claims depend on matched quality, an explicit participant-level estimand, a target-specific control, and a downstream test that is attempted only after the representation change is verified.

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